Thomas Male

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Professional Profile

Virtual Reality (VR) enthusiast and a driven Game Programmer. I have experience designing and developing applications using a variety of programming languages, including C++, C#, and OpenGL. My experience has enabled a strong understanding of programming practices and game design principles that deliver projects to a high-quality standard.

Most recently, my role as the Lead XR Developer at ByteEd has enabled me to build and refine a range of programming and development skills, including Mobile App Development, Extended Reality (XR), User Management, Database Development, and more.

The majority of the projects I have worked on utilized the Unity Game Engine. However, I have also exercised my skills in several different formats, including Console Applications (C# and C++), Web development, and compatibility layers.

I have participated in training to become an accredited personal learning and development (PLD) provider. This credential has been used in a number of schools and tertiary education facilities where I provided PLD for both students and teachers, in either XR Development or C# programming. These educational facilities were located in New Zealand, the UAE, and Malaysia.

Education and Qualifications

Bachelor of Software Engineering – Game Programming

Media Design School

February 2016 – November 2018
Courses

- *Game Design Principles*
- *Advanced Software Engineering and Programming for Games*
- *Rapid Game Prototyping*
- *Artificial Intelligence*
- *Technology Leverage for Games*
- *Software Engineering Principles and Practices*
- *Game Engine Development*
- *Software Engineering Capstone Project*

A and AS level Subjects (Cambridge)

ACG Strathallan

February 2007 – December 2015
Subjects

- *A2 Mathematics with Statistics*
- *AS Physics*
- *AS Computing*
- *AS English Language*

Relevant Experience

11 / 2020 – Present XR Development Team Lead, ByteEd, Auckland

Team leader of the XR Development team at ByteEd. This includes developing and launching multiple Virtual and Augmented Reality projects. I also managed and aided in the development of educational materials for students and teachers covering digital technologies and XR project creation. The development of most of our projects used the Unity Game Engine, which used C# as the main programming language. This process included designing, prototyping, testing, iterating, and developing.

As team leader, I was responsible for managing other developers on our XR projects. This includes managing contract workers, part-time developers, or university students who worked with us for work experience. To manage the several different projects as well as different developers, I kept a project management and collaboration tool up to date that each team member could access.

As the lead developer, I was also in charge of all the other digital-related projects that were being developed or maintained by ByteEd. This included multiple Mobile Applications, Custom website development, Database development and maintenance, AI tool integration, research and development, and more. This involved facilitating communication with clients, determining priority of features, testing, and signing off on updates ready to be sent out or launched.

11 / 2020 – Present XR Development PLD Provider, ByteEd, Auckland

NZ NCEA Accredited PLD provider for XR Development and digital literacy. While working at ByteEd, I also provided numerous PLD's to schools and tertiary education. The primary focus of these PLD's was to teach XR Development or

digital literacy, to help teachers better understand the topics and introduce VR technology into their school.

I have provided PLD nationally and internationally in the United Arab Emirates (UAE) and Malaysia. When providing these PLD courses, we tailored each one to the school's needs. This process involved meetings with both the teachers and team leaders prior to the PLD facilitation, discussing recommended topics and software to use for each individual case. I was responsible for facilitating this communication.

Associated Experience

05 / 2019 – 11 / 2020 Customer Service Representative, VRVOOM, Auckland

As a customer service representative I look after customers from when they walk in the door, guiding them around the VR Center and handling transactions. I teach customers how to use the various stations and equipment inside the store while also starting the experiences for them. I am in charge of maintaining and fixing any technical issues where possible, and cleaning the store after large events / closing up. Recently learned how to make and serve coffee with a barista-grade machine.

10 / 2018 – 10 / 2018 Armageddon Booth Attendant (Volunteer Work), Media Design School

I was stationed at the Media Design School booth at Armageddon expo 2018. My responsibilities included guiding people through the Virtual Reality experience and answering any questions they had about Media Design School. I also helped set up and look after the equipment while it was being used.

Other Experience

06 / 2015 – 08 / 2015 Dog Caretaker, Brookby Lodge Kennels and Cattery
I worked in the Kennels, looking after the dogs and their environment. My responsibilities included cleaning kennels, preparing meals for the dogs, and supervising the dogs to keep them entertained and ensure a safe environment. This helped develop my teamwork skills and showed that I am a responsible, hardworking, and reliable person.

04 / 2016 – 06 / 2016 Computer Hot Staging, Advanced Retail
I hot-staged computers, set them up, and made them ready to be shipped to retail companies such as Paper Plus. My responsibilities included setting up the OS on a "blank" computer, formatting hard drives, installing retail software, and configuring them. This helped develop my IT and Computer skills.

05 / 2017 – 01 / 2019 Casual / On Call Labourer, Turanga Creek Vineyard
I worked as a casual laborer helping with general work around the vineyard. My responsibilities included pruning vines, wire pulling, and placing large netting across the vineyard which took a group of 6 people. This helped develop my teamwork and communication skills

11 / 2017 – 02 / 2018 Casual Contract Labourer, A S Wilcox & Sons Limited
I worked under a casual contract, helping with general labor. My responsibilities included working on a harvester line and working with other interns harvesting seedlings to be planted. This helped me improve my communication skills,

Projects

Play Code Learn - Mobile App by ByteEd 09 / 2021 – Present

I was the lead XR Developer on this project. I was in charge of designing and developing the majority of the application, from prototyping, testing, iterations, and launching.

- Augmented Reality (AR) app that brings to life the physical STEM kit, also developed by ByteEd. This uses the mobile device's camera and Unity's XR system to track AR Anchors and place digital objects into a physical space.
- Created to facilitate and aid young learners in understanding the fundamentals of digital technologies and literacy.
- This app uses many different systems, such as: Online Database and authentication, AR Tracking and model technology, Account / Licence-based access, and in-app coding and simulation.

AlphaBee - Mobile App by ByteEd & Stylefit (*Unlaunched*) 08 / 2023 – 10 / 2025

I was the lead XR Developer on this project. I was in charge of designing and developing the majority of the application, from prototyping, testing, and iterating based on client feedback. This App was developed with the intent to be a standalone package that would be marketed and released on the ByteEd platform, using the same AR Technology.

- Augmented Reality (AR) app that brings to life the physical STEM kit, also developed by ByteEd. This uses the mobile device's camera and Unity's XR system to track AR Anchors and place digital objects into a physical space.
- Created to facilitate and aid young learners in understanding, speaking, and writing in the English language. Aimed at students learning to read and write at schools.
- Included in the App is a number of exercises relating to “Say it”, “Write it”, and “Story it”. As they progress through the App, the activities get harder and more challenging as they progress through each of the phonemes and graphemes of the English language, following the NCEA format.
- This app has a custom writing activity, where students must trace over a letter and get a score based on how accurate and smooth their letter is.

WriteWhizz - Mobile App by ByteEd & EdConnect (*Awaiting Launch)* 06 / 2024 – Present

I was the lead XR Developer on this project. I was in charge of designing and developing the majority of the application. I also liaised with the Client directly, giving regular updates, taking feedback, and iterating based on the changes needed.

- Artificial Intelligence (AI) powered app that brings a personal learning assistant and learning activities to each student. WriteWhizz focuses on teaching students how to write the English alphabet.
- Has a custom-made grading algorithm to calculate an accuracy score for the written letter, while also being able to detect common mistakes.
- Connects to an AI-powered pipeline that provides custom and unique verbal feedback to students in the role of a personal assistant, based on the exact letter formation provided by the student.
- Included with the App is a website portal that allows teachers to login with their credentials and fetch details about their

students. They can view their students progress, assign activities, manage their students credentials, and download custom QR Codes that allow students to quickly log in to their account using an authentication pipeline from the App.
- This app uses many different systems, such as: Online Database and authentication, custom AI pipeline, Account / Licence-based access, In-app purchases, and validation.

ByteEd Website **01 / 2025 – Present**

I was the lead XR Developer on this project. I coordinated directly with the contracted developers to help understand the scope of the project and what features needed to be added. The ByteEd website is also a teaching portal that allows teachers to log in and access education-related content that was created by the ByteEd Team.

This educational content needed to be sorted within a database, and have permissions set up to allow only certain access to areas within the website. Users could then view the content directly within the website using a PDF viewing layer, but it also enabled users to download the PDF for use within their classroom.

After the contract was finished, I became the main developer for all remaining updates and features on the website. This included:

- Adding additional courses to the platform
- Updating course information
- Adding in working user management, such as password recovery and the ability to log in with social accounts (e.g., Gmail and Microsoft)
- Setting up the mailing service and DNS to automate emails from our domain when password resets were requested or automated messages needed to be sent
- Adding Student Enrollment features so that teachers could view, manage, and enroll new students into the platform under their account, given that they had enough seats available
- Automatic QR-code generation for each student that could be printed and used as a login method in conjunction with our ByteEd App.

Earthquake Preparedness simulation - VReady **03 / 2021 – 01/2022**

I was the lead XR Developer on this project. I was in charge of designing and developing the majority of the application. I also liaised with the Client directly, giving regular updates, taking feedback, and iterating based on the changes needed.

- Virtual Reality (VR) simulation of an earthquake within a school.
- Created together with the team at VReady to be as accurate and educational as possible to teach others what to do in an earthquake.
- My focus of the development is on Virtual Reality and interactions within the game. Selecting UI options, navigating through an environment, and triggering events within the simulation.

5G R&D VR Streaming Project - ByteEd **09 / 2021 – 12/2022**

Research and Development project experimenting with streaming VR content from a dedicated PC to a 5G-enabled mobile device. The main goal of this research and development was to test the capabilities of the 5G network launch in NZ and test if VR streaming could be achieved over a 5G satellite network.

Various prototypes were created and tested. The most successful one included not only streaming the VR content to the mobile device, but also streaming back digital input from both the mobile device's gyroscope and a Bluetooth controller, and mapping the digital inputs to VR controller inputs. The difficulty here was that we needed to be able to register the inputs in any VR game, so we used a custom-made compatibility software that spoke between the mobile device and the SteamVR instance, which it turn then spoke to any given game that was being launched.

Smith & Squire (University Project - Alpha) **04 / 2018 – 11/2018**
- Virtual Reality (VR) Asymmetrical Local Co-op Crafting Game. The game runs off a single computer with one player in VR and the other player using the mouse and keyboard.
- This game was chosen to be showcased at the Media Design School booth at Armageddon Expo Auckland 2018.
- My focus of the development is the Virtual Reality aspect of the game. This includes: Physical Interactions, Dynamic crafting systems, and VR Locomotion.

Gravity Tester **07 / 2018 – 10 / 2018**
- Mobile Casual Game. Dodge flying debris and stay alive for as long as possible.
- The focus of this project was practicing porting to a range of devices. So far, we've Successfully ported to the PS Vita & Android platforms. We have also added in Ads, In-app Purchases, Achievements and Social Integration with Twitter and Facebook for the Android Platform.

VR UI Made Easy (Unreleased Unity Plugin) **03 / 2018 – 06 / 2018**
- A Unity Plugin that allows you to easily create diegetic User Interface (UI) for Virtual Reality (VR). This plugin consists of only scripts as it uses Unity's base UI elements to create diegetic UI.

Research Paper on Combating Motion Sickness in Virtual Reality Headsets **02 / 2018 – 06 / 2018**
- A Research paper created to study the causes of motion sickness when using Virtual Reality Head Mounted Displays (VR

HMD). I also tested and recorded various methods on how to reduce the feeling of motion sickness and how well they rated against each other.
- This paper was then used (with my permission) to further enhance my lecturers research paper about cyber sickness in HMD's

Ghost Roast – Global Game Jam 2019 **01 / 2019**
- I attended the Global Game Jam 2019 (GGJ19) and created a game with a team of 6 within 48 hours. This was a simple game where you must hunt down ghosts floating around you house. The ghosts are invisible and are only visible when you get close enough to them. The ghosts move and float objects around as they move so you have some form of spotting them.

Programming Languages

- Unity / C# - 8 Years of Experience
- C++ - 3 Years of Experience
- SQL - 3 Years of Experience
- Open GL - 1 Year of Experience
- Lua, MySQL, HTML, XML -1 Year of Experience

Key Skills

- C++ Programming
- Unity C#
- Mathematics
- Conflict Resolution
- OpenGL
- Project Management
- Adaptability
- VR Development

Other / Personal Details

Interests include:
- Virtual Reality, Video Games, Board Games, Design and Developing Games, Reading Books, 3D Printing and modelling
- Exercising / Gym, Archery, Walking / Hiking

REFERENCES ARE AVAILABLE ON REQUEST